
Detection of free sulfhydryl groups

Introduction

Ellman's assay for detection of free sulfhydryl groups on Sumo-tagged peptides expressed in E coli:

This protocol was optimised from available protocols at: <https://www.bmglabtech.com/en/application-notes/ellmans-assay-for-in-solution-quantification-of-sulfhydryl-groups/>

Ellman's reagent (5,5'-dithio-bis-(2-nitrobenzoic acid) or DTNB), produces a yellow-colored product when it reacts with sulfhydryl groups. This yellow-colored product can be quantified via measurement of the absorbance at 408- 412 nm.

DTNB reacts with a free sulfhydryl group to yield a mixed disulfide and 2-nitro-5-thiobenzoic acid. TNB is the "colored" species produced in this reaction and has a high molar extinction coefficient in the visible range. The molar extinction coefficient of TNB is reported to be 13,600 M⁻¹cm⁻¹ at 412 nm and pH 8.0.

Materials



- › Reaction Buffer (RB): 0.15 M Tris-HCl pH 8.0, Approximately 100-150 ml per plate to be analysed
- › Cysteine Hydrochloride Monohydrate (MW 175.6)
- › Ellman's Reagent 5,5-dithio-bis-(2-nitrobenzoic acid)
- › 96 well plate (recommended: Greiner UV-STAR Microplate, 96 well, half area, clear. Ref: 675801)



Procedure

Prepare Working solutions:

1. Fill up a template for the plate layout (you can use this one, S = Standard, U = Unknowns, B = Blanks). NOTE: Use the same labels for replicates as this is necessary for them to be recognised as replicates during the analysis.

Plate Layout Sample



	1	2	3	4	5	6	7	8	9	10	11	12
A	S1	S1	S1	U1	U1	U9	U9	U17	U17	U25	U25	U33
B	S2	S2	S2	U2	U2	U10	U10	U18	U18	U26	U26	U33
C	S3	S3	S3	U3	U3	U11	U11	U19	U19	U27	U27	U34
D	S4	S4	S4	U4	U4	U12	U12	U20	U20	U28	U28	U34
E	S5	S5	S5	U5	U5	U13	U13	U21	U21	U29	U29	B
F	S6	S6	S6	U6	U6	U14	U14	U22	U22	U30	U30	B
G	S7	S7	S7	U7	U7	U15	U15	U23	U23	U31	U31	B
H	S8	S8	S8	U8	U8	U16	U16	U24	U24	U32	U32	B

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2. **Prepare Buffer:** Use a buffer like Tris-HCl adjusted to a pH ensuring the stability and solubility of thiol groups in the protein. Appro
 3. **Prepare DTNB stock solution:** Make 10mM DTNB stock solution by dissolving 40mg DTNB in 10ml DMSO. The stock solution can be stored at 4°C for 3 months.
 4. **Prepare DTNB Solution:** Dilute the stock solution 2000 fold with 0.1M Tris-HCl pH 8 to make 5 µM DTNB working solution. = 250 ul of Stock DTNB in 50 ml of Tris-HCl
 5. **Prepare DTNB "Standards" solution:** Dilute the stock solution 1000 fold with 0.1M Tris-HCl pH 8 to make 5 µM DTNB working solution. = 125 ul of Stock DTNB in 50 ml of Tris-HCl.

Prepare standards:

6. Dissolve Cysteine in the DTNB "Standards" buffer solution and making serial dilutions with the buffer as specified on Table 1. Preparation of Standard Solutions

Table1. Preparation of Standard Solutions



	A	B	C	D	E	F	G	H	I
1	Standard	Concentration (mM)	Cysteine (ml)	Cysteine Vol (uL)	Reaction Buffer (mL)	Reaction Buffer (uL)	Final Vol (ml)	STD 1 (ul)	Concentration (uM)
2	Stock	10.000	17.564mg		10.000	10000	10.0		
3	Dilution 1	1.000	0.1	100	0.900	900	1.0	100	1000
4	Dilution 2	0.100	0.1	100	0.900	900	1.0	100	100
5	STD 1	0.050	0.05	50	0.950	950	1.0	50	50
6	STD 2	0.040	0.04	40	0.960	960	1.0	40	40
7	STD 3	0.030	0.03	30	0.970	970	1.0	30	30
8	STD 4	0.020	0.02	20	0.980	980	1.0	20	20
9	STD 5	0.010	0.01	10	0.990	990	1.0	10	10
10	STD 6	0.005	0.005	5	0.995	995	1.0	5	5
11	STD 7	0.003	0.0025	2.5	0.998	997.5	1.0	2.5	2.5
12	STD 8	0.001	0.001	1	0.999	999	1.0	0	1
13	SUM			357.5		8642.5		255	

Load samples and standards on the plate:

7. Load 50 μ l of DTNB "Standard" reaction buffer onto the wells that are going to be loaded with standards
8. Load 50 μ l of DTNB solution onto the wells that are going to be loaded with unknowns
9. Load 50 μ l of each standards onto a plate using replicates (preferably triplicates) for each standard.
10. Load 50 μ l of each sample on the corresponding well (preferably in duplicates)
11. Incubate samples for 15 min at room temperature on an orbital platform shaker

Perform the analysis:

12. Read the samples on the plate reader at MoES performing a spectral scanning preferable from 300-500 nm. Define to read in horizontal scanning format (for speed).
13. Copy the raw reads onto an Excel file and save it on your corresponding /Expdata folder
14. Export the plate layout as CSV and save it as plate_layout.csv [to be loaded on the analysis jupyternotebook) and load it onto the same folder used for the analysis.
15. Generate a table with Standards and standards concentration (using same labels as in plate_layout.csv), as in table below and load it onto the same folder used for the analysis.

Standard concentrations (uM) sample

	A	B
1	Standard	uM_Concentration
2	S1	50
3	S2	40
4	S3	30
5	S4	20
6	S5	10
7	S6	5
8	S7	2.5
9	S8	1
10	B	0

16. Preferably, make a folder (within your directory) with the 3 files needed for the analysis (plate layout, standards and raw plate reader data) and copy the following notebook to the folder:

`/mnt/home/yensifb/software/ds_expression/Ellman/Ellmans_analysis.ipynb`

17. Open the notebook and modify the paths to the files as specified on the # File paths cell and run the notebook making note of the Max absorbance and standards that may not fit the curve or trial other curve models included at the end of the notebook.